Shailaja Singh

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SKILLS

Programming: C++, Python, JavaScript, HTML/CSS, SQL, C, Java, Kotlin(Android)
Tools: Android Studio, IntelliJ, PyCharm, Clion, Postman, Git, Bootstrap, Agile

PROJECTS

Sofia Chess Engine *Nov 2025 – Mar 2026*

- Engineered a fully UCI-compliant chess engine ("Sofia") in C++17; The engine communicates with any standard GUI (Arena, Cute Chess, etc.) via the UCI protocol.
- Implemented alpha-beta negamax with iterative deepening & quiescence + 5 move-ordering heuristics, achieving 85%+ fail-high-first and depth 6 search <1s on mid-game positions.
- Parallelized perft move-tree using C++ std::async (1 thread/root move + bulk-counting at depths 1–3), boosting throughput 10× (2–4M → 20–24M nodes/sec).
- Github: github.com/sshailaja03/Chess-Engine

Revizo – Gamified Learning App *Aug 2025 – Oct 2025*

Team Lead

- Implemented SM-2 spaced repetition in Kotlin to schedule 40+ flashcards across 5 decks, dynamically updating ease factors (1.3–2.5) and intervals
- Architected Android app using MVVM + Jetpack (Room, Live Data, Coroutines) with a 6-table schema and repository pattern; enabled fully async, lifecycle-safe data flow across 4 activities, reducing UI–DB coupling
- Built real-time battle mode using coroutines + reactive state (Live Data) with 15s/question timer; captured accuracy, response time, and XP to power post-session analytics and performance insights
- Github: github.com/sshailaja03/Revizo-Learning-App

Python File Encryptor *Jan 2025 – Apr 2025*

- Implemented AES-128 Fernet encryption using Python cryptography, enabling secure encryption/decryption of any file type with 100% data unreadability without key
- Built a 2-command CLI workflow using argparse, reducing user effort to single-command use with zero configuration
- Developed a <70 LOC lightweight utility with 1 dependency, ensuring minimal footprint and easy deployment.
- Github: github.com/sshailaja03/file-encryptor

ACTIVITIES AND LEADERSHIP

Lovely Professional University **Jalandhar, Punjab**
Teaching Assistant and Counselor for 2029 Batch (C++ and UI&UX Designing) *Aug 2025 – Current*

Optimus Student Organisation, LPU **Jalandhar, Punjab**
Vice - President *Nov 2024 – Current*

Leader of an organization of 100+ members, hosting tech-focused events (25+) and focused on building a vibrant, informed, and active student tech community.

CERTIFICATIONS

DSA Summer Bootcamp (LPU) | *July 2025*
Cloud Computing (NPTEL) | *March 2025*

EDUCATION

Lovely Professional University **Jalandhar, Punjab**
B. Tech in Computer Science *Expected Graduation, June 2027*
Concentrations: Cyber Security and UI UX Designing
CGPA: 7.62, *Dean's List*

Related Coursework: Data Structures & Algorithms, Combinatorics, Cyber Security, UI & UX Designer, Object-Oriented Programming, Statistics & Applications